(b) Values of r and T are given for different moons in Table 2.1.

Table 2.1

moon	r / 10^8 m	T / 10^3 s	$\lg(r/10^8\,\text{m})$	$\lg(T/10^3\,\text{s})$
Pandora	1.42	52 ± 5		
Mimas	1.86	81 ± 5		
Enceladus	2.38	120 ± 10		
Tethys	2.95	170 ± 10		
Dione	3.77	240 ± 20		
Rhea	5.28	390 ± 30		

Calculate and record values of $\lg(r/10^8\,\text{m})$ and $\lg(T/10^3\,\text{s})$ in Table 2.1. Include the absolute uncertainties in $\lg(T/10^3\,\text{s})$. [2]

(c) **(i)** Plot a graph of $\lg(T/10^3\,\text{s})$ against $\lg(r/10^8\,\text{m})$. Include error bars for $\lg(T/10^3\,\text{s})$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]